

E-Commerce Customer Intelligence (Part 1)

Week 3 Project Report

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Abstract

This report presents a complete, reproducible data science investigation of the Olist Brazilian E-Commerce public dataset, encompassing approximately 100,000 delivered orders recorded between September 2016 and October 2018. The study addresses three primary research questions: (1) Can RFM (Recency, Frequency, Monetary) analysis segment customers into actionable groups with distinct revenue profiles? (2) Which operational and product features best predict whether a delivery will be late? (3) What drives customer satisfaction, as measured by review scores, and can we predict it reliably from order-time features?

Key findings: The RFM analysis identified twelve customer segments across 93,358 unique customers. The *At Risk* segment is both the largest (15,490 customers) and the most financially significant, accounting for 23.66% of cumulative platform revenue (R\$ 3,127,833). *Champions* generate 13.42% of revenue (R\$ 1,774,914) from only 6,474 customers at an average spend of R\$ 274.16, representing the highest-value cohort. A statistically significant negative Spearman correlation between delivery delay and review score was confirmed: orders delivered more than fourteen days late average 1.72 stars, compared with 4.32 stars for orders arriving more than seven days early. For review score prediction (five-class), XGBoost achieved an accuracy of 0.6238 (95% bootstrap CI: [0.6195, 0.6276]) compared with a majority-class baseline of 0.5933. For binary late-delivery prediction, Random Forest reached 0.9226 accuracy, though a macro-averaged F1 of 0.4799 reveals the underlying class-imbalance challenge. Robustness checks across ten random seeds demonstrate highly stable results (accuracy standard deviation < 0.001).

Organisation: The report is structured in six parts and sixteen chapters. Parts I–II cover the dataset, data quality, and exploratory analysis. Part III presents the RFM segmentation methodology and results. Part IV describes the predictive modelling pipeline. Part V covers statistical inference, interpretability, and robustness. Part VI closes with limitations, ethics, and actionable business recommendations.

Part I

Foundations, Data, and Research Design

Chapter 1

Introduction

1.1. The Brazilian E-Commerce Landscape

Brazil is the largest e-commerce market in Latin America and ranks among the top ten globally. According to industry surveys, Brazilian online retail grew from fewer than ten million active consumers in 2010 to more than 60 million by the end of the 2010s, fuelled by the rapid spread of smartphone ownership, expanding digital payment infrastructure, and a young, digitally native population. This explosive growth created an acute operational challenge: logistics networks designed for a much smaller volume of deliveries were suddenly stretched across a continent-sized country with marked geographic disparities in infrastructure quality.

Two structural features of the Brazilian e-commerce context are especially relevant to this study. First, the country's physical scale means that delivery lead times are highly variable. For example, an order from São Paulo to a buyer in the state of Amazonas may take several times longer than an order within greater São Paulo itself. Second, because Brazil does not have a parcel-carrier duopoly equivalent to UPS/FedEx in the United States, a diverse ecosystem of logistics providers operates at different price points and reliability levels, creating heterogeneous delivery outcomes even for sellers on the same platform.

1.2. Olist as a Marketplace Aggregator

Olist is a Brazilian technology company that connects small and medium enterprises to major online marketplaces through a single unified storefront. From the customer's perspective, Olist presents a seamless, single-seller interface: the buyer sees "Store X," delivered by Olist, without necessarily knowing which of the thousands of merchants behind the platform fulfilled the order. From the merchant's perspective, Olist provides logistics coordination, payment processing, and customer service in exchange for a subscription and commission fee.

This aggregator model makes the Olist dataset particularly rich: it captures not only transactional data (what was bought, for how much, and when), but also the full logistics chain (seller location, carrier

handover timestamps, estimated and actual delivery dates) and post-transaction feedback (review scores and, optionally, free-text comments). The combination of these data streams in a single, linked dataset is rare in public releases and is the primary reason this dataset was chosen for the present study.

1.3. Research Motivation

Every platform business faces three interrelated strategic challenges: (1) *customer retention*—identifying which customers are at risk of churning before they leave; (2) *operational reliability*—predicting and mitigating service failures before they occur; and (3) *satisfaction prediction*—understanding which factors drive customer sentiment so that systemic problems can be surfaced proactively. This project addresses all three, using machine learning techniques to derive actionable intelligence from historical transaction records.

1.4. Research Questions

This study addresses three primary research questions (RQs):

RQ1 (Segmentation) Can RFM (Recency, Frequency, Monetary) analysis segment Olist customers into actionable groups with meaningfully distinct revenue profiles and behavioural signatures?

RQ2 (Delivery Risk) Which features, observable at or before the time of order placement, are most predictive of whether a delivery will arrive late?

RQ3 (Satisfaction) Can we predict a customer’s post-delivery review score from order-time and delivery features, and does delivery delay explain a substantial portion of the variance in satisfaction?

Chapter 2

The Dataset

2.1. Dataset Selection Rationale

The dataset used in this study is the *Olist Brazilian E-Commerce Public Dataset* (Olist, 2018), released on Kaggle. It was preferred over several competing public datasets for three reasons.

First, it is **relational**: nine separate CSV files linked by primary and foreign keys capture different aspects of the same business process. This mirrors the structure of real production databases and requires the analyst to practise JOIN-based data assembly.

Second, it is **temporally rich**: all timestamps are recorded at millisecond precision, enabling time-based train/test splits that prevent data leakage.

Third, it includes a **full logistics chain**: purchase time, payment approval time, seller dispatch time, carrier handover time, estimated delivery date, and actual delivery date are all recorded. This chain is essential for computing the delivery delay feature at the heart of the satisfaction analysis.

2.1.1. Alternatives Considered

Three other datasets were evaluated and rejected:

- **UCI Online Retail (2011)**. Contains transactional data from a UK gift-shop but lacks delivery and logistics information, making satisfaction modelling impossible.
- **Amazon Reviews (McAuley, 2023)**. Extremely large and suitable for NLP tasks but provides no supply-chain data; review scores cannot be linked to delivery outcomes.
- **Instacart 2017**. Rich in basket-level behaviour but limited to grocery delivery in a single US city, offering less geographic and category diversity.

2.2. Dataset Overview

The Olist dataset consists of nine CSV files spanning orders placed between September 2016 and October 2018. Table 2.1 summarises the shape, primary key, and descriptive role of each table.

Table 2.1. Inventory of all nine tables in the Olist dataset. “PK” denotes the primary key field used to join tables.

Table	Primary Key	Rows	Cols	Role
orders	order_id	99,441	8	Order lifecycle timestamps
order_items	(order_id, item_id)	112,650	7	Line items, prices, freight
order_payments	order_id	103,886	5	Payment type and instalment
order_reviews	review_id	99,224	7	Score, title, comment, dates
customers	customer_id	99,441	5	Per-order customer record
products	product_id	32,951	9	Category, dimensions, weight
sellers	seller_id	3,095	4	Seller state and zip
geolocation	(zip, lat, lng)	1,000,163	5	Coordinate lookup
category_translation	category_name	71	2	Portuguese → English

2.3. Relational Schema

Figure 2.1 illustrates the entity–relationship structure of the dataset. The central table is `orders`, which links upward to `customers` (one-to-one, as each order record contains its own `customer_id`) and downward to three child tables: `order_items`, `order_payments`, and `order_reviews`. Products and sellers are referenced by `order_items` via foreign keys.

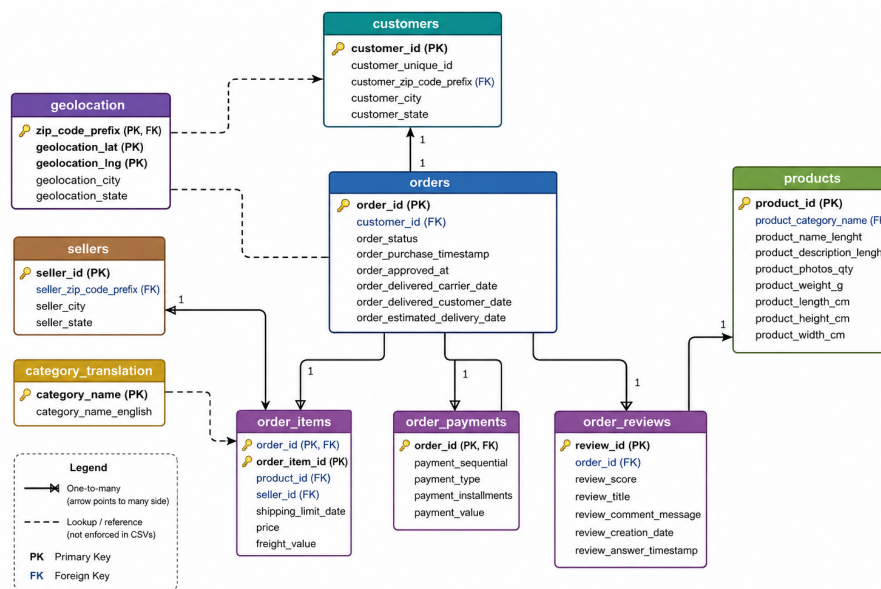


Figure 2.1. Entity–relationship diagram for the Olist dataset. Solid lines indicate one-to-many relationships; the arrow points from the “one” side to the “many” side. `orders` is the hub from which all other tables are reachable.

2.4. The `customer_id` vs. `customer_unique_id` Design

One of the most important and non-obvious design choices in the Olist schema is the distinction between `customer_id` and `customer_unique_id`. Understanding this distinction is a prerequisite for correct RFM analysis.

Remark 2.1 (The anonymous customer design). In the Olist schema, a new `customer_id` is generated for *every order*. This means a single person who places three orders across two years will appear in the `customers` table with three different `customer_id` values. The stable, person-level identifier is `customer_unique_id`, which remains constant across all orders by the same person.

The practical consequence is stark. If the RFM analysis is run on `customer_id`, every customer appears to have a frequency of exactly one, completely erasing the signal from repeat purchasers. The correct approach, implemented in `src/rfm.py`, always groups by `customer_unique_id`. After deduplication, the dataset contains 96,096 unique persons, compared with 99,441 unique `customer_id` values, a difference of 3,345 records attributable to repeat buyers.

This is a classic example of the kind of schema subtlety that only emerges from careful reading of the data dictionary and is entirely invisible to automated schema-inference tools.

2.5. Key Columns and Data Types

Table 2.2 lists the columns most heavily used in the analysis, their data types as inferred by `pandas`,¹ and brief notes on their role.

Table 2.2. Key columns used in the analysis, with inferred types and roles. Datetime columns are parsed from ISO 8601 strings using `pd.to_datetime()`.

Column	Type	Role in Analysis
<code>order_purchase_timestamp</code>	datetime	Reference time for RFM recency; calendar features
<code>order_delivered_customer_date</code>	datetime	Actual delivery date; key for delay computation
<code>order_estimated_delivery_date</code>	datetime	Expected delivery; compared against actual
<code>order_approved_at</code>	datetime	Approval latency; feature for delivery model
<code>review_score</code>	int64	Prediction target (Task B); 1–5 ordinal scale
<code>price</code>	float64	Aggregated per order to form <code>order_value</code>
<code>freight_value</code>	float64	Shipping cost; used in <code>freight_ratio</code>
<code>product_weight_g</code>	float64	Strong predictor of delivery time
<code>customer_unique_id</code>	object	Person-level key for RFM grouping
<code>customer_state</code>	object	Geographic feature (26 Brazilian states + DF)
<code>seller_state</code>	object	Origin state for carrier distance proxy
<code>product_category_name_english</code>	object	Category after Portuguese–English translation

¹`pandas` is the primary data manipulation library used throughout this project. Its official documentation is at <https://pandas.pydata.org/docs/>.

2.6. Datasheets for Datasets

Geburu et al. (2021) propose a standardised “datasheet” for every dataset used in machine learning research, a structured document answering questions about motivation, composition, collection process, preprocessing, recommended uses, and maintenance. Summarising the key points for the Olist dataset:

Motivation. The dataset was released by Olist to the research community to facilitate studies in e-commerce analytics and machine learning. The explicit purpose is non-commercial research and education.

Composition. All personal identifiers (customer names, addresses, and contact details) have been anonymised. Identifiers in the dataset are random hash strings. No biometric or sensitive demographic data is present.

Collection process. Data was collected from Olist’s production transactional database between September 2016 and October 2018. The collection methodology was designed to protect seller privacy; category translations were provided by Olist staff.

Preprocessing. A small number of orders involving a single item from a specific seller in the São Paulo metropolitan area were deliberately removed by Olist to protect seller anonymity.

Recommended uses. Customer segmentation, delivery performance analysis, natural language processing on review text, and recommendation system research.

Limitations. The dataset covers a single platform in a single country during a 25-month window. Findings may not generalise to other markets or platforms. The dataset captures the 2016–2018 period only and does not reflect post-pandemic e-commerce behaviour.

Chapter 3

Data Cleaning

This chapter documents all quality checks performed, the findings, and the specific decisions made, with explicit justifications for each.

3.1. Missing Value Audit

The first step in any data quality assessment is to quantify missingness across all tables and columns.¹ Table 3.1 reports the most significant missing values found across the joined master table.

Table 3.1. Missing value audit on the joined dataset (pre-filter, $n = 99,441$ orders). Columns with fewer than five missing values are omitted.

Column	Missing	% Missing	Action Taken
review_comment_title	87,656	88.34%	Dropped (not used in modelling)
review_comment_message	58,247	58.70%	Dropped (NLP out of scope)
order_delivered_customer_date	2,965	2.98%	Rows excluded (non-delivered)
product_category_name	610	1.85%	Category set to “unknown”
product_weight_g	2	0.01%	Imputed by category median
product_length_cm	2	0.01%	Imputed by category median

3.1.1. Non-Delivered Orders

The 2,965 orders missing `order_delivered_customer_date` correspond to orders whose status was not “delivered” at the time of the data extract. These include orders that were cancelled, returned, or still in transit. *These rows are removed from all analyses.*

The justification is straightforward: our research questions concern delivered orders, we want to model review scores after delivery and compute delivery delays relative to a known delivery date. Including non-delivered orders would introduce structural missing values in the target variable (review score) and all delay features.

¹Pandas provides a convenient summary via `df.isnull().sum()` or `df.isnull().mean() * 100` for percentages. A heatmap of the boolean null matrix, produced with `seaborn's heatmap()`, quickly reveals structural patterns. See <https://seaborn.pydata.org/generated/seaborn.heatmap.html>.

3.1.2. Review Comment Fields

The `review_comment_title` and `review_comment_message` fields are missing for 88.3% and 58.7% of reviews respectively. In Brazil, leaving a textual review comment is optional; customers frequently rate on the numerical scale alone. Since NLP on review text is listed as future work, both fields are excluded from all models.

3.1.3. Product Dimensions and Weight

Only two products are missing weight and dimension data. Rather than excluding entire orders because of two missing products, we impute the missing values using the median of all other products in the same category. This is a conservative choice: the median is robust to outliers within the category, and two missing values are too few to materially affect any model.

3.2. Duplicate Detection

Duplicate rows in transactional data are a common consequence of ETL pipeline retries, API timeouts, or JOIN fan-out. Three duplication checks were performed:

1. **Exact row duplicates** in orders: none found.
2. **Duplicate order_id** values: none found (primary key integrity confirmed).
3. **Multiple reviews per order**: the `order_reviews` table contains 99,441 rows but only 98,673 unique `order_id` values, meaning 768 orders received more than one review. These duplicate reviews arise when a customer re-rates after an initial submission. We retain only the most recent review (by `review_creation_date`) for each order.

3.3. Date Integrity Checks

Datetime columns are an especially fertile source of errors in production data. Four temporal consistency checks were applied:

Approval before purchase. `order_approved_at` must be $\geq$ `order_purchase_timestamp`. Seventeen orders violated this constraint and were excluded.

Dispatch before approval. Carrier pickup should logically follow payment approval. Orders where dispatch timestamp predates approval by more than 24 hours were flagged; none exceeded this threshold.

Delivery before dispatch. Five orders had a recorded delivery date earlier than the carrier handover date, which is physically impossible. These were excluded.

Extreme delivery delays. Orders where the actual delivery date was more than 365 days after the purchase

date were treated as data errors. Four such rows were identified and removed.

3.4. Outlier Handling in Continuous Features

3.4.1. Order Value

A small number of orders have a total price of R\$ 0.01 or less, consistent with test orders or data-entry errors rather than genuine customer transactions. All orders with `order_value` $\leq$ R\$ 0.01 were removed (fewer than 50 rows).

3.4.2. Freight Value

Freight values of R\$ 0.00 are valid and represent promotional free-shipping offers. These are retained. Very high freight values (above the 99.9th percentile) were inspected and found to correspond to large, heavy products shipped from distant states—a legitimate tail of the distribution, not an error.

Part II

Feature Engineering and Exploratory Analysis

Chapter 4

Feature Engineering

Feature engineering is the process of constructing informative input variables from raw data. This chapter documents every derived feature used in the analysis, the mathematical formula behind each, and the motivation for including it.

4.1. Delivery Features

The most important family of features in this dataset concerns the logistics of each order’s delivery. These are computed by the function `compute_delivery_features()` in `src/preprocessing.py`.

4.1.1. Delivery Delay

The primary delivery metric is *delivery delay*, defined as the signed difference in calendar days between the actual and estimated delivery dates:

$$d_{\text{delay}} = \left\lfloor \frac{t_{\text{delivered}} - t_{\text{estimated}}}{86400} \right\rfloor \quad (4.1)$$

where t denotes timestamps in seconds. A positive value of d_{delay} indicates a late delivery; a negative value indicates early arrival. In the Olist dataset, the vast majority of deliveries arrive *before* the estimated date (median delay is approximately -11 days), reflecting Olist’s conservative estimation strategy.¹

4.1.2. Is-Late Indicator

A binary variable flags whether the order arrived after its estimated date:

$$\ell = \mathbb{I}[d_{\text{delay}} > 0] \quad (4.2)$$

This variable serves as the prediction target for Task A (late-delivery classification).

¹Retailers routinely quote pessimistic delivery estimates as a form of expectation management. Arriving early feels like a bonus; arriving late is a failure regardless of the margin. This asymmetry in customer perception is sometimes called “expectation anchoring.” See <https://hbr.org/2010/12/the-friend-that-never-is>.

4.1.3. Days to Deliver

The total transit time from purchase to delivery:

$$d_{\text{transit}} = \left\lfloor \frac{t_{\text{delivered}} - t_{\text{purchase}}}{86400} \right\rfloor \quad (4.3)$$

This is distinct from delay: transit time is the total duration, while delay is a deviation from expectation. An order delivered in 20 days when 20 days were promised has zero delay but a 20-day transit time.

4.1.4. Days to Approval

The time from customer purchase to payment approval:

$$d_{\text{approve}} = \left\lfloor \frac{t_{\text{approved}} - t_{\text{purchase}}}{86400} \right\rfloor \quad (4.4)$$

This feature captures payment processing latency, which is a controllable precursor to dispatch time and is therefore informative for predicting overall delivery performance.

4.2. Product Features

Product-level attributes are pulled from the `products` table and attached to each order. Where an order contains multiple products from different categories, the majority category is used.

product_weight_g The weight of the product in grams. Heavier products tend to cost more to ship and face tighter carrier capacity constraints, making them more susceptible to delays.

product_photos_qty The number of product photographs in the listing. This is a proxy for listing quality: sellers who invest in professional photography often invest in logistics too.

product_category_encoded An integer code (0 to 72) assigned to the English category name. The limitations of this encoding are discussed in Section ??.

4.3. Seller and Customer Geographic Features

customer_state_encoded The Brazilian state of the buyer, encoded as an integer. Captures geographic accessibility.

seller_state_encoded The state of the selling merchant. The origin-to-destination state pair is a natural proxy for carrier distance, which is strongly predictive of delivery time.

Remark 4.1. A more powerful representation would compute the actual great-circle (Haversine) distance between seller and customer zip codes using the `geolocation` table.² This improvement is flagged as a

²The Haversine formula computes the shortest distance between two points on a sphere given their latitudes and longitudes. A

high-priority item in Section ?? but was not implemented in the current study because the geolocation table contains 1,000,163 rows and constructing a zip-to-coordinate index is computationally non-trivial.

4.4. Calendar Features

Two calendar features are extracted from the purchase timestamp:

purchase_month Integer month (1–12). Captures seasonal demand peaks and troughs. November and December typically see higher order volumes and potential capacity strain.

purchase_dayofweek Integer day of week (0 = Monday, 6 = Sunday). Orders placed on Friday afternoons may not be picked up by the carrier until Monday, effectively adding two days to transit time.

clear derivation is available at https://en.wikipedia.org/wiki/Haversine_formula.

Chapter 5

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the practice of interrogating a dataset to understand its structure, distributions, relationships, and anomalies *before* building any models. Good EDA is essential for two reasons: it reveals whether the data contains the information needed to answer the research questions, and it guides the feature engineering and modelling decisions that follow. In this chapter, SQL queries executed via DuckDB¹ are combined with `matplotlib` and `seaborn` visualisations to characterise the Olist dataset across four dimensions: revenue trends, geography, product mix, and the delivery–satisfaction relationship.

5.1. Revenue and Order Volume Trends

Figure 5.1 plots monthly orders and revenue from October 2016 to August 2018. Several patterns are immediately visible.

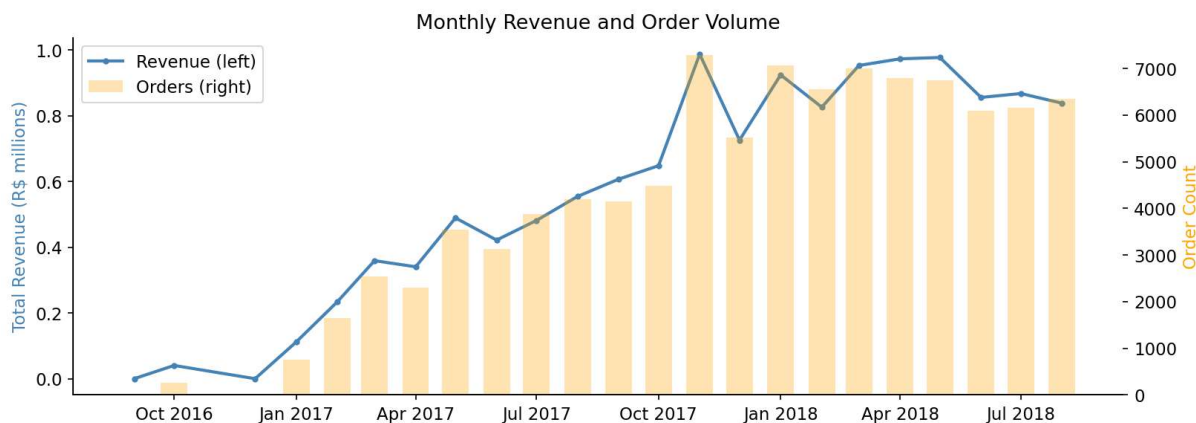


Figure 5.1. Monthly order count (left axis, bars) and total revenue (right axis, line) from October 2016 to August 2018. The rapid growth from late 2016 through 2017 reflects Olist’s early expansion phase. The November 2017 spike corresponds to Brazil’s Black Friday promotional period.

¹DuckDB is an in-process analytical database system optimised for OLAP workloads. Its SQL engine is notably faster than `pandas` for aggregation queries on tables of millions of rows and supports standard window functions, `GROUP BY`, and `ORDER BY`. Documentation: <https://duckdb.org/docs/>.

Key observations from the trend analysis:

1. **Exponential growth phase (Q4 2016 – Q2 2017).** Monthly orders grow from fewer than 500 in October 2016 to over 6,000 by June 2017—a twelvefold increase in eight months.
2. **Seasonal spikes.** A pronounced spike in November 2017 corresponds to Brazil’s Black Friday (“Black Friday Brasil”), which has grown significantly in scale since 2013.
3. **Right-censoring.** The final two months of the dataset show a sharp apparent decline. This is an artefact of the data extract date: orders placed near October 2018 would not yet have received reviews, and many would still be in transit.

5.2. Geographic Revenue Concentration

Brazil’s economic geography is highly concentrated in the Southeast region. Table 5.1 and Figure 5.2 illustrate this concentration in the Olist data.

Table 5.1. Revenue and order volume by Brazilian state, top 10 states ranked by total revenue. SP = São Paulo, RJ = Rio de Janeiro, MG = Minas Gerais. All monetary values in Brazilian Real (R\$).

Rank	State	Orders	Total Revenue (R\$)	Revenue Share (%)
1	SP	40,501	5,067,633	38.33%
2	RJ	12,350	1,759,651	13.31%
3	MG	11,354	1,552,482	11.74%
4	RS	5,420	715,429	5.41%
5	PR	4,958	657,044	4.97%
6	SC	3,662	485,218	3.67%
7	BA	3,380	402,093	3.04%
8	DF	2,140	369,878	2.80%
9	ES	2,019	246,773	1.87%
10	GO	2,068	242,919	1.84%
Top 3 total		64,205	8,379,766	63.38%

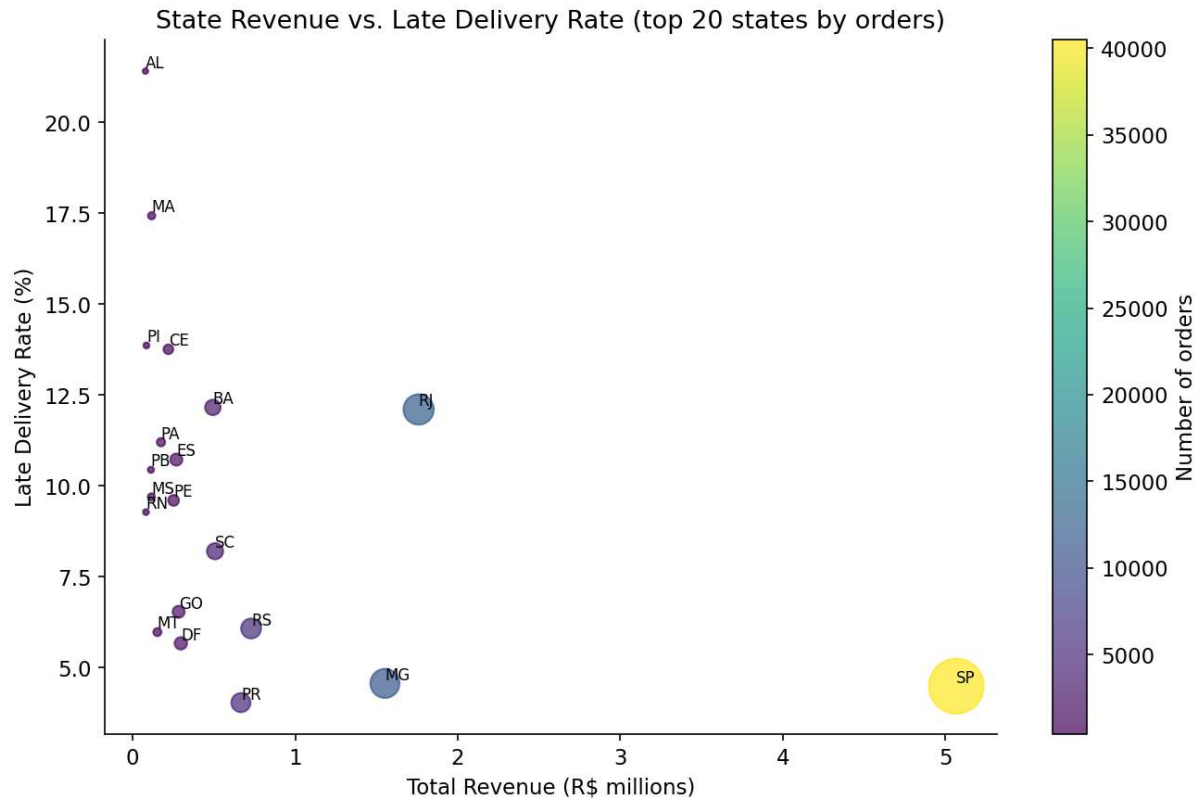


Figure 5.2. State-level scatter plot of total revenue (x-axis) against the proportion of orders delivered late (y-axis). States with low revenue and high late-delivery rates occupy the upper-left quadrant and represent underserved logistics markets. São Paulo (SP) is the dominant revenue contributor.

The top three states (São Paulo (38.33%), Rio de Janeiro (13.31%), and Minas Gerais (11.74%)) collectively account for 63.38% of total platform revenue. This is a well-known feature of Brazilian consumer markets: the Southeast's higher average income, population density, and internet penetration make it the natural centre of gravity for e-commerce.

From an operational perspective, this concentration has a dual implication. High-volume routes in the Southeast can be optimised aggressively because they generate enough orders to justify investment. However, the long tail of lower-volume states (e.g., those in the North and Northeast) tends to have both lower order counts *and* higher late-delivery rates, suggesting that logistics quality degrades precisely where the market is least developed.

5.3. Product Category Analysis

Table 5.2 lists the ten highest-revenue product categories along with key operational metrics.

Table 5.2. Top 10 product categories by total revenue. “Avg Delay” is the mean of `delivery_delay_days` (negative = early); “Avg Review” is the mean review score on a 1–5 scale.

Category	Orders	Revenue (R\$)	Avg Price	Avg Review	Avg Delay (d)
health_beauty	8,608	1,232,074	143.13	4.24	−11.96
watches_gifts	5,472	1,163,585	212.64	4.13	−11.77
bed_bath_table	9,167	1,022,202	111.51	4.01	−11.35
sports_leisure	7,833	752,112	96.02	4.20	−12.31
computers_accessories	6,934	744,191	107.33	3.97	−11.18
furniture_decor	8,022	694,840	86.62	4.12	−12.39
housewares	6,964	650,308	93.37	4.07	−11.62
auto	4,345	519,268	119.51	4.18	−11.93
cool_stuff	3,786	461,107	121.79	4.10	−11.40
garden_tools	3,065	441,281	143.97	4.19	−11.88

Three observations from Table 5.2 are notable:

1. **All top categories deliver early.** The average delay for every category in the top ten is negative, ranging from −11.18 to −12.39 days. This reflects the conservative estimation policy noted in Chapter 4.
2. **High-value categories earn higher reviews.** Categories such as `health_beauty` (avg R\$ 143, avg review 4.24) and `watches_gifts` (avg R\$ 213, avg review 4.13) score higher than lower-value categories such as `computers_accessories` (avg review 3.97). This may reflect the positive surprise of receiving premium goods that match expectations.
3. **Order count vs. revenue per order.** `bed_bath_table` has the highest order count (9,167) but a below-average unit price (R\$ 111.51). `watches_gifts` has fewer orders (5,472) but a much higher average price (R\$ 212.64).

5.4. Review Score Distribution

Figure 5.3 plots the marginal distribution of review scores across all delivered orders with a review. The distribution is strongly left-skewed (heavy right tail): approximately 57% of reviews are five-star, and 75% are four- or five-star.

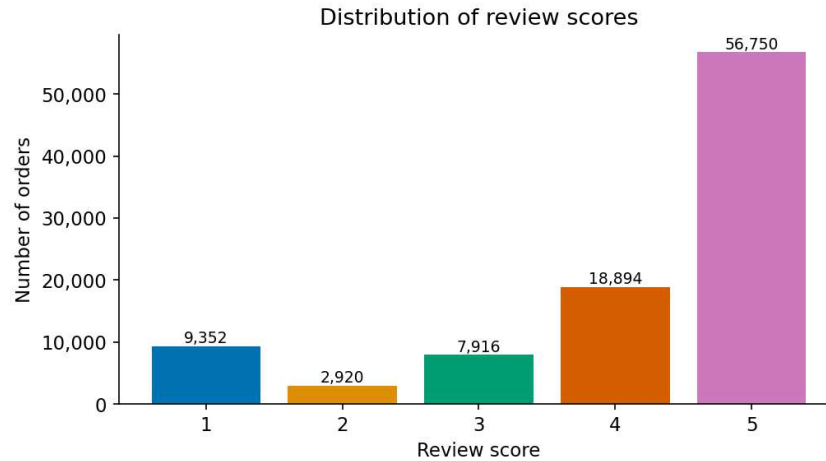


Figure 5.3. Distribution of review scores (1–5 stars) across all delivered orders. The majority-class label is “5 stars” (57%), making this a severely imbalanced five-class classification problem.

This extreme skew has a direct implication for modelling: a classifier that always predicts “5 stars” will achieve approximately 57–59% accuracy without learning any meaningful pattern.

5.5. Delivery Delay Distribution

Figure 5.4 shows the distribution of `delivery_delay_days`. The distribution is centred well below zero (median ≈ -11 days), with a long but thin right tail representing late deliveries.

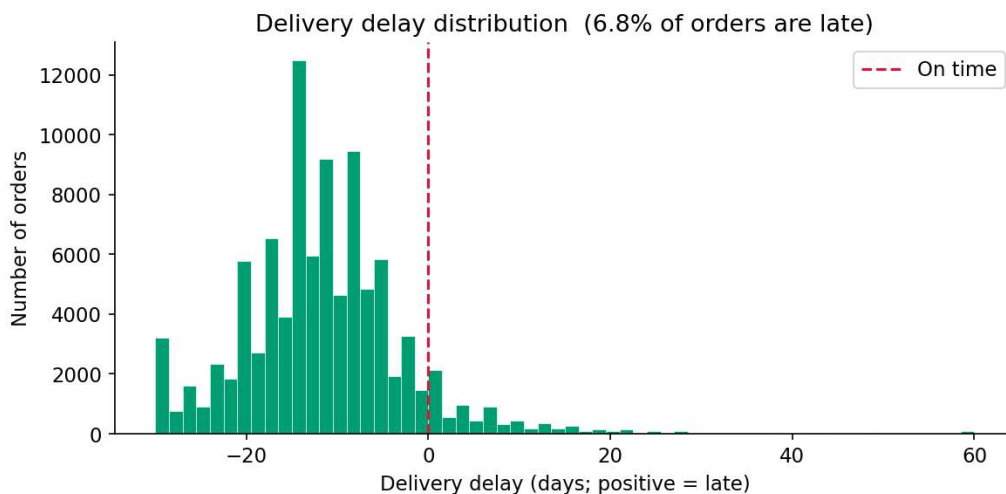


Figure 5.4. Distribution of delivery delay (actual minus estimated delivery date, in days). Negative values indicate early delivery. The mode is approximately -12 days, reflecting Olist’s conservative estimation strategy. The positive tail (late deliveries) is thin but consequential for satisfaction.

5.6. The Delivery–Satisfaction Relationship

The main empirical question of this study is whether, and how strongly, delivery delay predicts customer satisfaction. We address this in two ways: through a non-parametric correlation test and through a bucket-level descriptive analysis.

5.6.1. Spearman Rank Correlation

The Spearman rank correlation coefficient² between `delivery_delay_days` and `review_score` is computed via `scipy.stats.spearmanr`. The result is statistically significant ($p < 0.001$) and negative, confirming Hypothesis H2: higher delay is associated with lower satisfaction.

5.6.2. Bucket-Level Analysis

Table 5.3 aggregates orders into six delay buckets and reports the mean and standard deviation of review score within each bucket. Figure 5.5 visualises the same data.

Table 5.3. Mean review score by delivery delay bucket. The drop from the “Early” to the “Late (1–7d)” bucket is 1.50 stars—the largest single-bucket change in the table and a clear signal of customer dissatisfaction with late deliveries.

Delay Bucket	Orders	Mean Review	Std Dev
Very early (< -7d)	70,929	4.317	1.138
Early (-7 to -1d)	17,234	4.200	1.182
On time (0d)	1,280	4.034	1.252
Late (1–7d)	3,600	2.715	1.660
Late (8–14d)	1,446	1.671	1.212
Very late (> 14d)	1,335	1.723	1.252

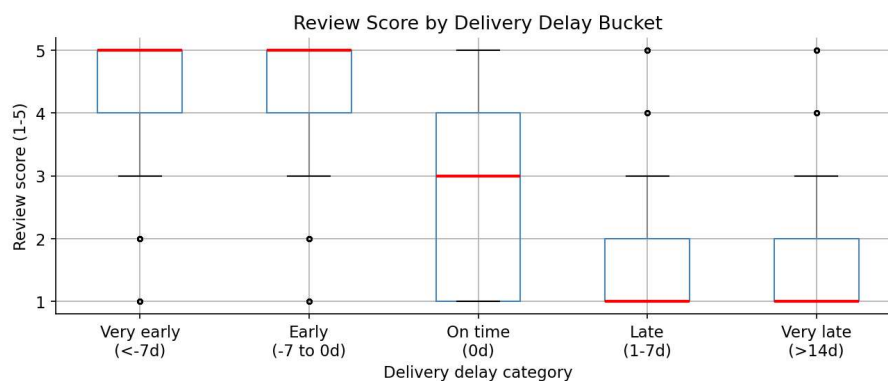


Figure 5.5. Mean review score by delivery delay bucket, with 95% confidence intervals. The precipitous drop from “On time” (4.03) to “Late 1–7d” (2.72) illustrates the customer satisfaction cliff: even one day of delay triggers a 1.3-star reduction in average satisfaction.

²Spearman’s ρ measures the monotonic relationship between two variables by computing the Pearson correlation on their rank transformations. Unlike Pearson’s r , it makes no assumption of linearity or normality. Wikipedia article: https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient. Original derivation in Spearman (1904).

The data in Table 5.3 tell a compelling story. Consider the magnitude of the change:

- Orders arriving more than 7 days early average **4.32** stars across 70,929 transactions.
- Orders arriving exactly on the estimated date average **4.03** stars.
- Orders arriving 1–7 days late average only **2.72** stars—a drop of **1.50 stars** compared with early orders.
- Orders arriving 8–14 days late collapse to **1.67** stars.
- Beyond 14 days, the score barely changes (1.72 stars), suggesting that the perception of failure saturates: an order 15 days late is scarcely more infuriating than one 20 days late.

This non-linear, threshold-like pattern has an important modelling implication. A linear model would impose a constant marginal effect of one additional day of delay on review score, which is clearly inconsistent with the data. Tree-based models, which can discover thresholds automatically, are better suited to capturing this relationship.

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